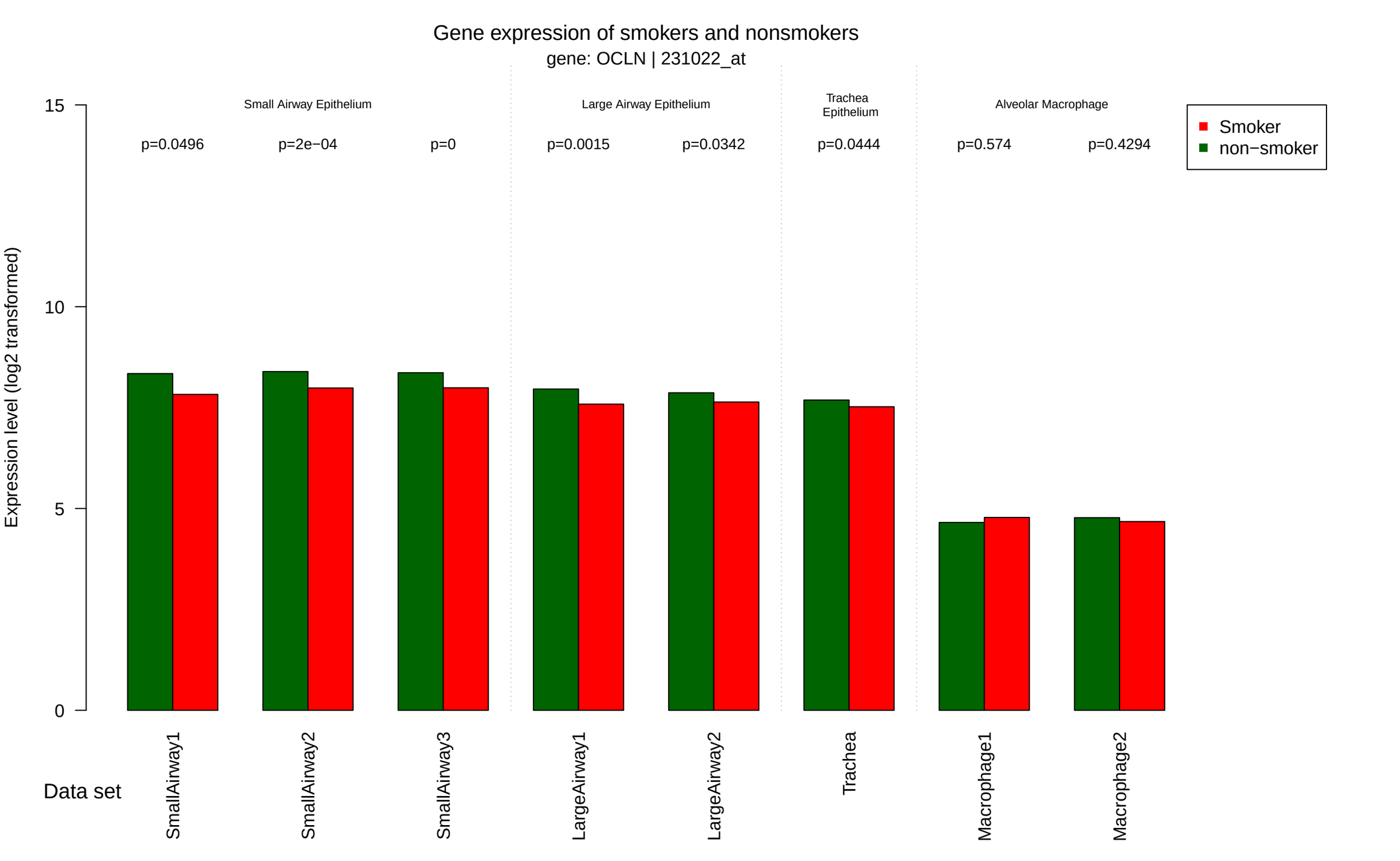
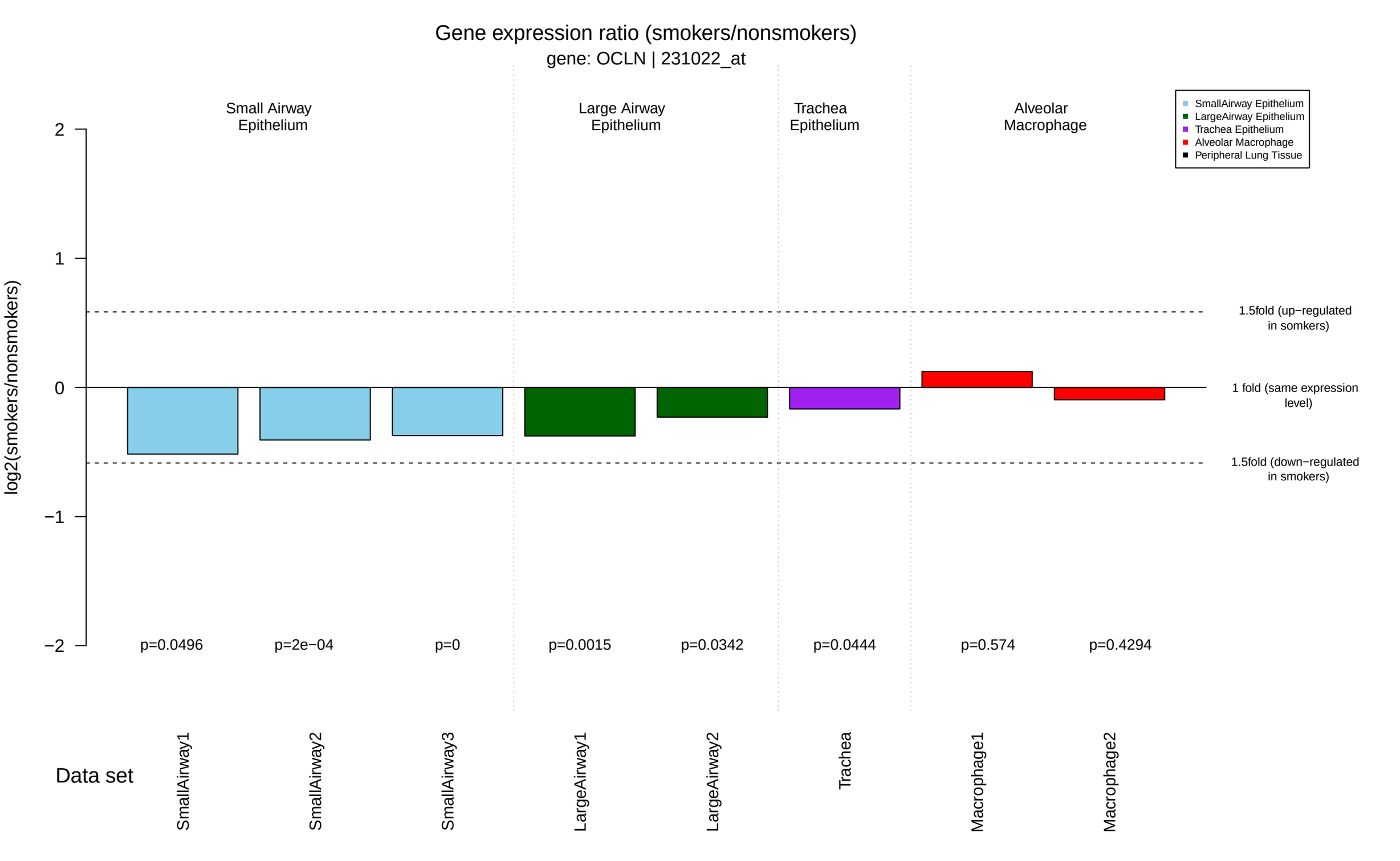
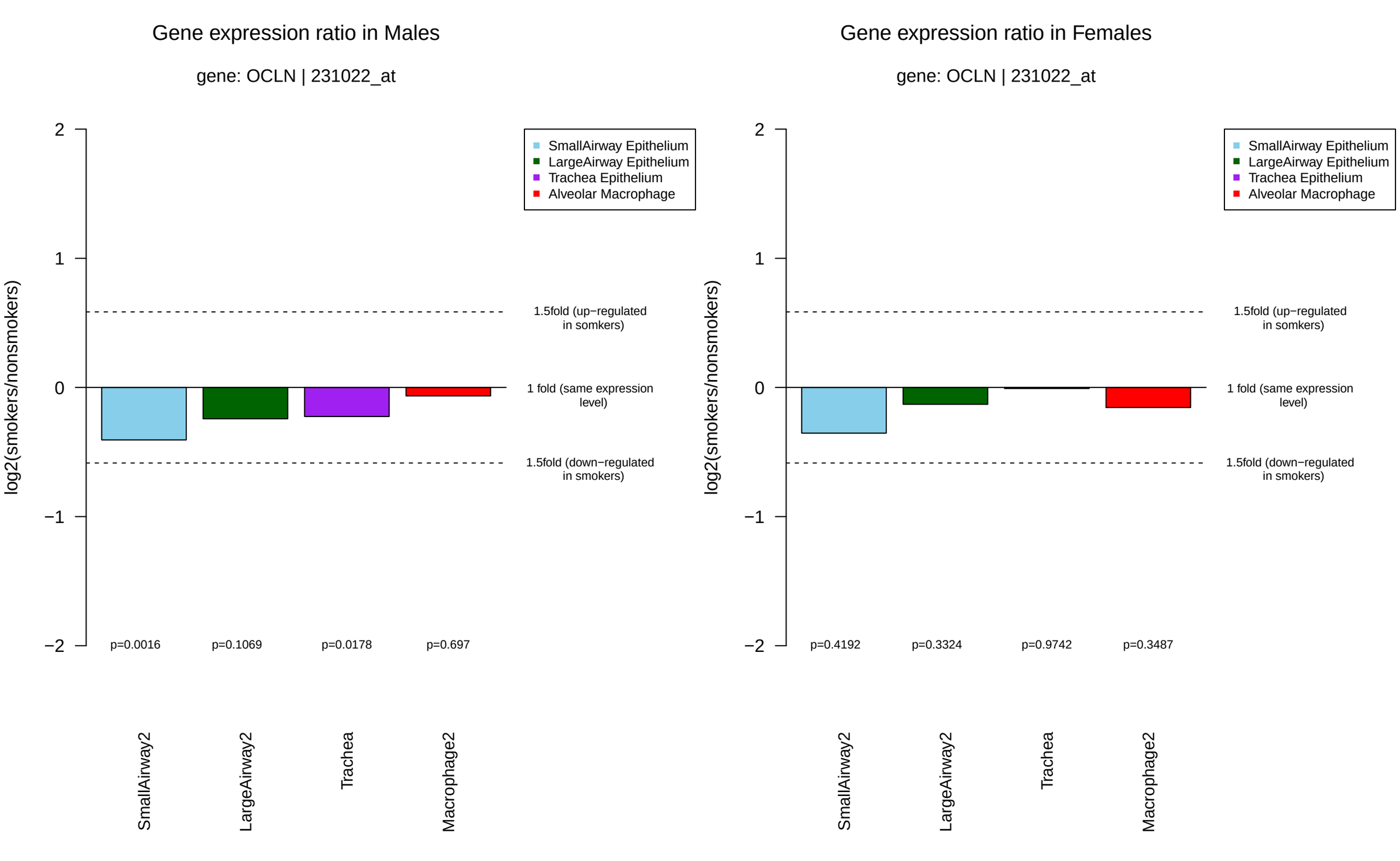
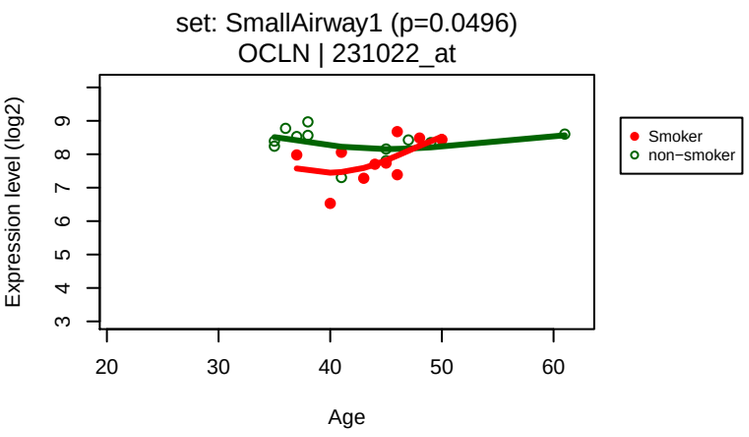
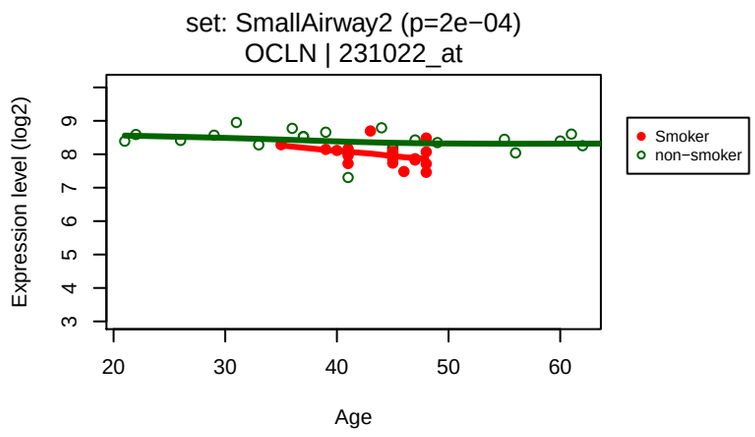
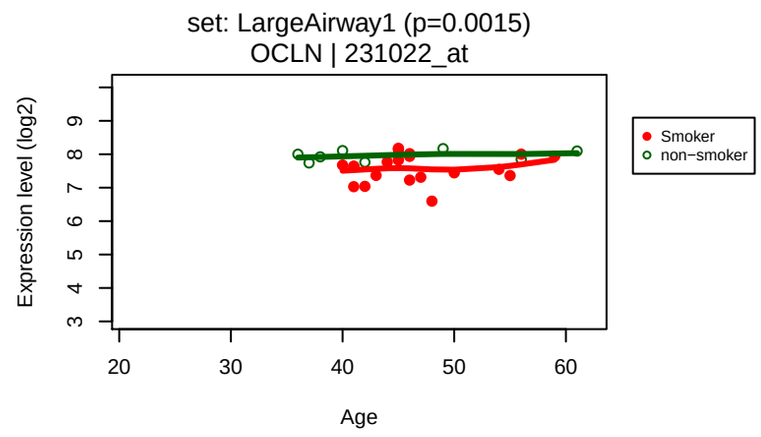

Gene expression of smokers and nonsmokers
gene: OCLN | 231022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0496
p=2e−04
p=0
p=0.0015
p=0.0342
p=0.0444
p=0.574
p=0.4294
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

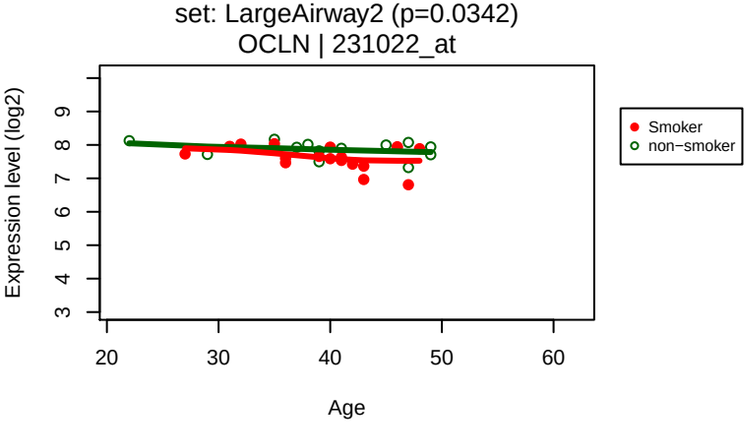

Gene expression ratio (smokers/nonsmokers)
gene: OCLN | 231022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0496
p=2e−04
p=0
p=0.0015
p=0.0342
p=0.0444
p=0.574
p=0.4294
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

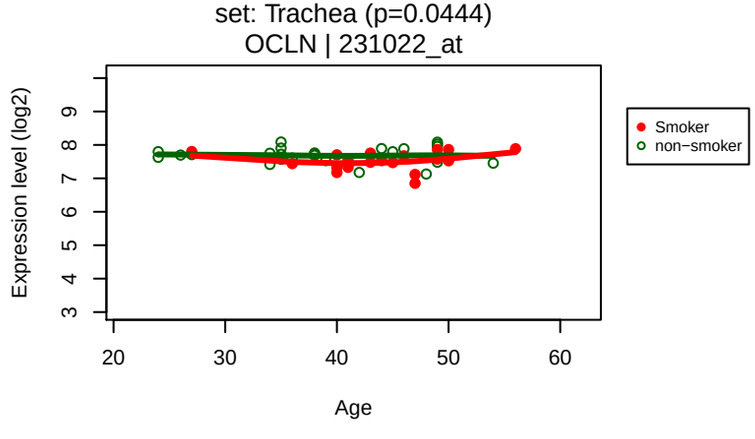

Gene expression ratio in Males
gene: OCLN | 231022_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0016
p=0.1069
p=0.0178
p=0.697
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OCLN | 231022_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4192
p=0.3324
p=0.9742
p=0.3487
SmallAirway2
LargeAirway2
Trachea
Macrophage2

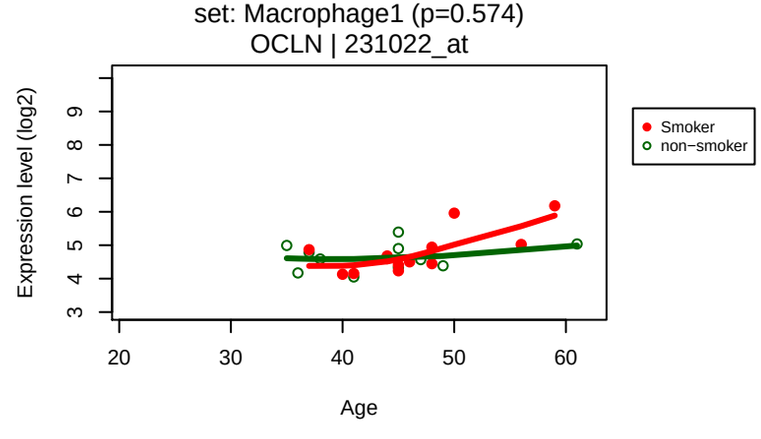

set: SmallAirway1 (p=0.0496)
OCLN | 231022_at
Expression level (log2)
Age
Smoker
non−smoker

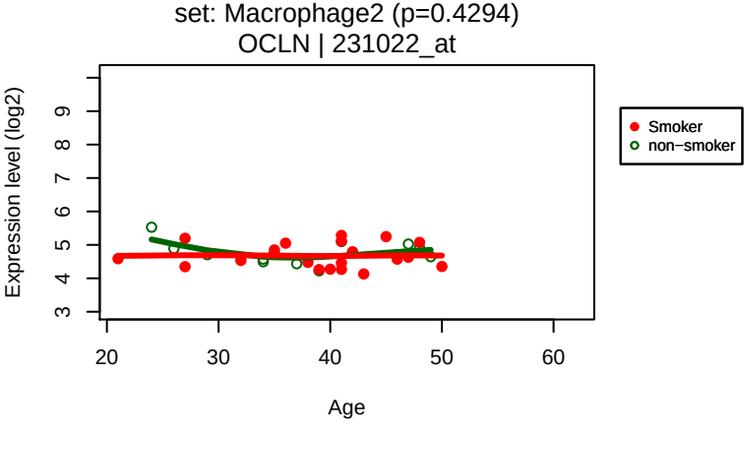

set: SmallAirway2 (p=2e−04)
OCLN | 231022_at
Expression level (log2)
Age
Smoker
non−smoker

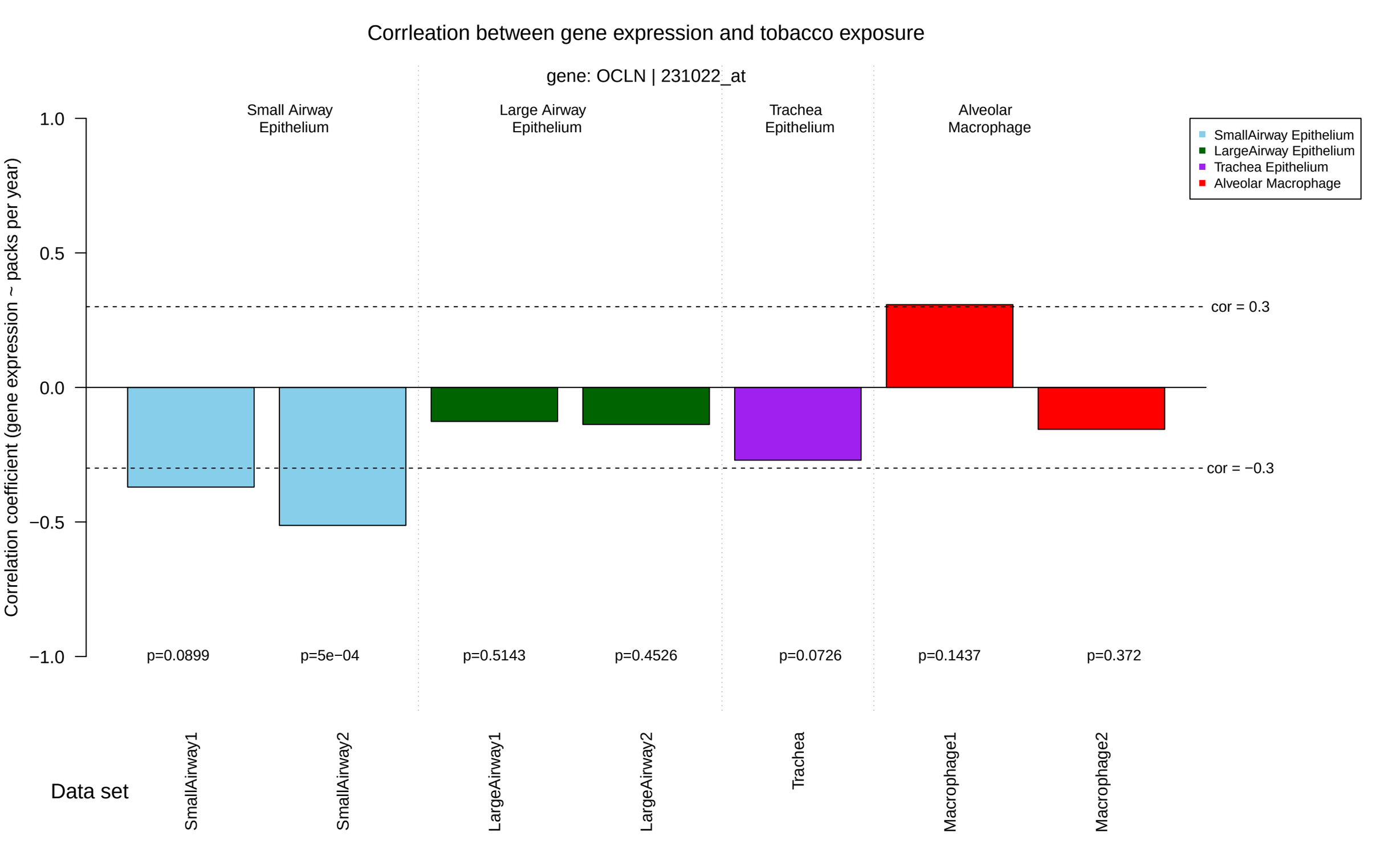

set: LargeAirway1 (p=0.0015)
OCLN | 231022_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0342)
OCLN | 231022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0444)
OCLN | 231022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.574)
OCLN | 231022_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4294)
OCLN | 231022_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OCLN | 231022_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0899
p=5e−04
p=0.5143
p=0.4526
p=0.0726
p=0.1437
p=0.372
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2